Data Imbalance Testing and Handling System

CODE: 26-1-D-3

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What Is Data Imbalance?

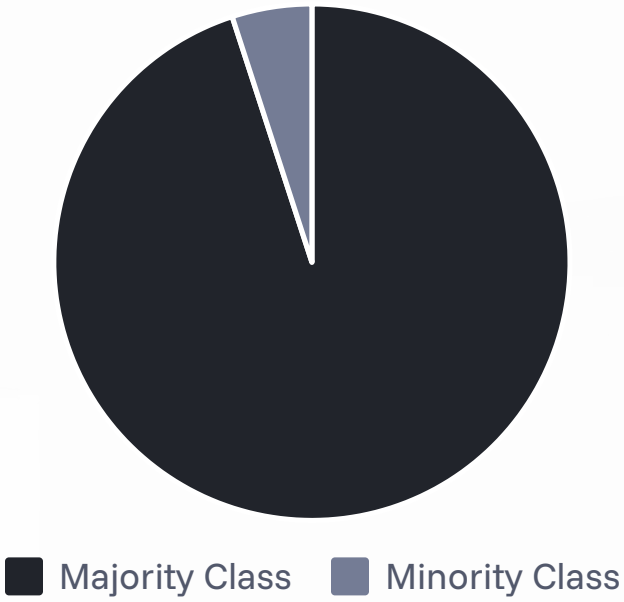
When one class appears far more often than others in the dataset.

Many real datasets are uneven: most samples belong to one "majority" class.

The minority class can be rare but important (e.g., fraud, faults, disease).

A model can look accurate by mostly predicting the majority class.

Therefore we must detect imbalance and evaluate with the right metrics.



ABSTRACT

Problem

- Real-world datasets frequently suffer from class imbalance.
- Critical minority classes are underrepresented.
- Leads to misleadingly high model accuracy.
- Masks severe failures in crucial, rare cases (fraud detection, medical diagnoses).

Challenge

- Existing solutions are fragmented.
- Poor documentation and inconsistent methodologies.
- Difficult to reproduce results.
- Hard to reliably compare different handling techniques.

Our Solution

- Unified system for diagnosing data imbalance.
- Implementing handling strategies.
- Comparing results effectively.
- Generating comprehensive documentation.

Real-World Impact

Class Imbalance

Skewed data leads to biased models.

Performance Risk

Models fail on rare, critical cases.

Consequences

Affects fraud, medical diagnosis, fairness.

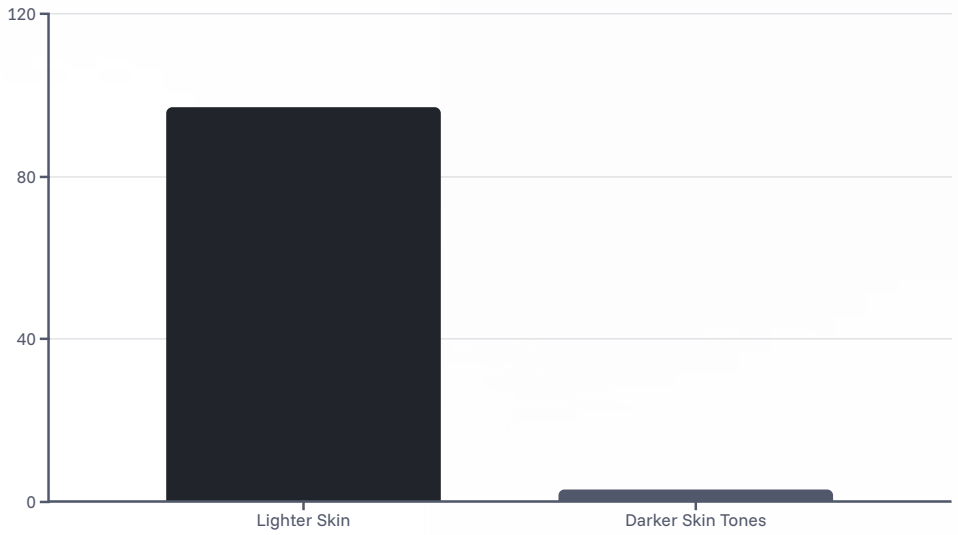
Real-World Example: Melanoma Detection Bias

- **Imbalanced medical datasets can make models less reliable for minority groups, especially in high-stakes diagnosis.**
- **In dermatology AI, severe data skew can lead to systematic bias.**
- **Models can perform well on lighter skin, while frequently failing to recognize life-threatening conditions on darker skin tones.**

97%

Melanoma prediction models at high risk of bias due to severe data skew (Kleinberg et al., 2022).

Performance by Skin Tone



The Accuracy Paradox: Why Accuracy Misleads

Consider the following confusion matrix for an imbalanced dataset:

	Predicted Positive	Predicted Negative
Actual Positive	TP: 0	FN: 50
Actual Negative	FP: 0	TN: 4950

The accuracy for this model is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{0 + 4950}{0 + 4950 + 0 + 50} = \frac{4950}{5000} = 0.99$$

- 99% accuracy achieved by ignoring minority class entirely
- Perfect accuracy on majority, complete failure on minority
- Demonstrates why accuracy alone is insufficient

Diagnosing Imbalance: Metrics & Pitfalls

Key Diagnostic Metrics

Imbalance Ratio (IR)

Measures class skew severity.

$$IR = \frac{N_{maj}}{N_{min}}$$

Higher IR = more severe imbalance.

G-Mean (Geometric Mean)

Balances sensitivity & specificity.

$$G - Mean = \sqrt{TPR \times TNR}$$

TPR = True Positive Rate (Recall/Sensitivity)

TNR = True Negative Rate (Specificity)

Low G-Mean indicates majority class bias.

IR shows how severe the class skew is, and G-Mean checks whether performance is balanced across both classes.

Without them, accuracy can look “good” while the model is biased toward the majority class.

Two Main Families of Solutions

Data-Level Approach (External Method)

Modify dataset before training to reduce class imbalance.

Algorithm-Level Approach (Internal Method)

Adjust the learning algorithm to prioritize the minority class .

Project Focus: Study both approaches for a transparent, reproducible workflow.

Data-Level Methods: Balancing Through Resampling

- **Modify dataset before training to reduce class imbalance**
- **Two main approaches: Over-sampling and Under-sampling**

Over-sampling

Increase minority class samples

Under-sampling

Reduce majority class samples

OVER-SAMPLING

SMOTE: Synthetic Minority Over-sampling

How SMOTE Works

01

Select minority sample

02

Find k-nearest neighbors

03

Generate synthetic samples via interpolation

Benefits

- Increases minority class without duplication
- Adds variation via interpolation

Risks

- Risk of overgeneralization
- Can cause class overlap

NearMiss: Distance-Based Under-Sampling

Strategically selects majority samples using distance rules.

Key Variants:

- NearMiss-1: Keeps majority samples closest to minority neighbors
- NearMiss-2: Uses average distance to farthest minority neighbors
- NearMiss-3: For each minority sample, keeps closest majority samples
- Most Distant: selects majority samples whose average distances to the three nearest minority samples are the largest

Trade-off:

- Deleting data discards potentially valuable information

Existing Tools & Software

Imbalanced-learn

- Scikit-Learn Integration
- Comprehensive Resampling Library
- High Academic Reliability

SMOTE-variants (Python Package)

- 85 over-sampling methods
- Multi-Class & Selection
- Research Specialized

WEKA

- No-Code Interface
- Dual Strategies
- Extensible

KEEL

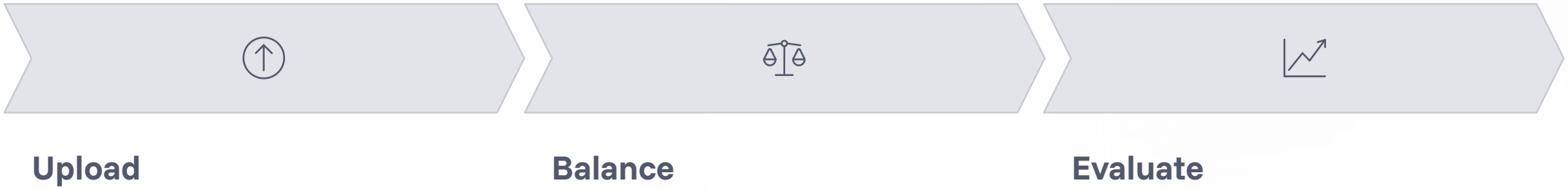
- Standardized Benchmarking Repository
- Specialized Imbalanced Module
- Scientific Validation Tools

Identified Gap: Existing tools lack integrated workflow for diagnosis, handling, evaluation, and documentation.

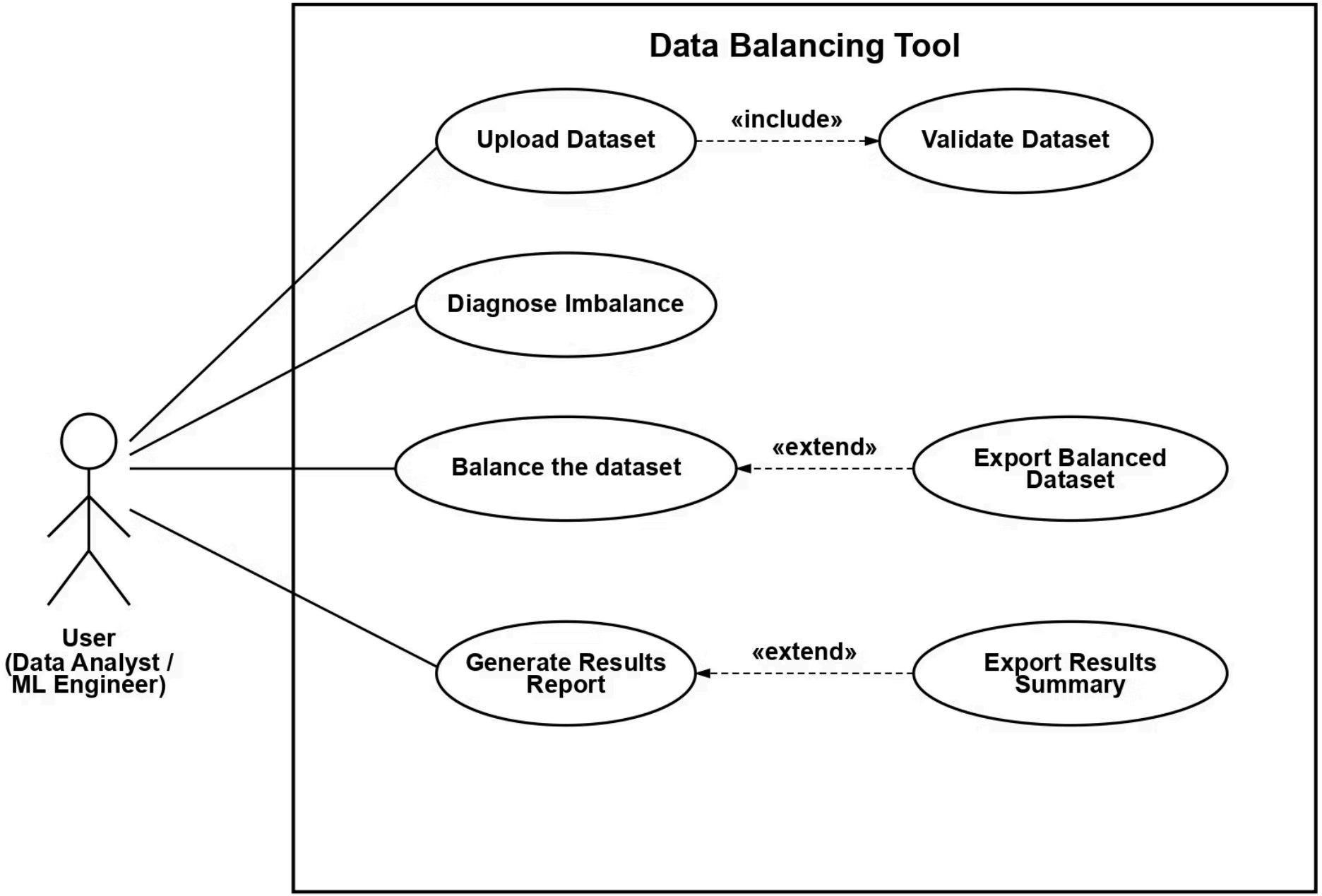
From Methods to System Usage

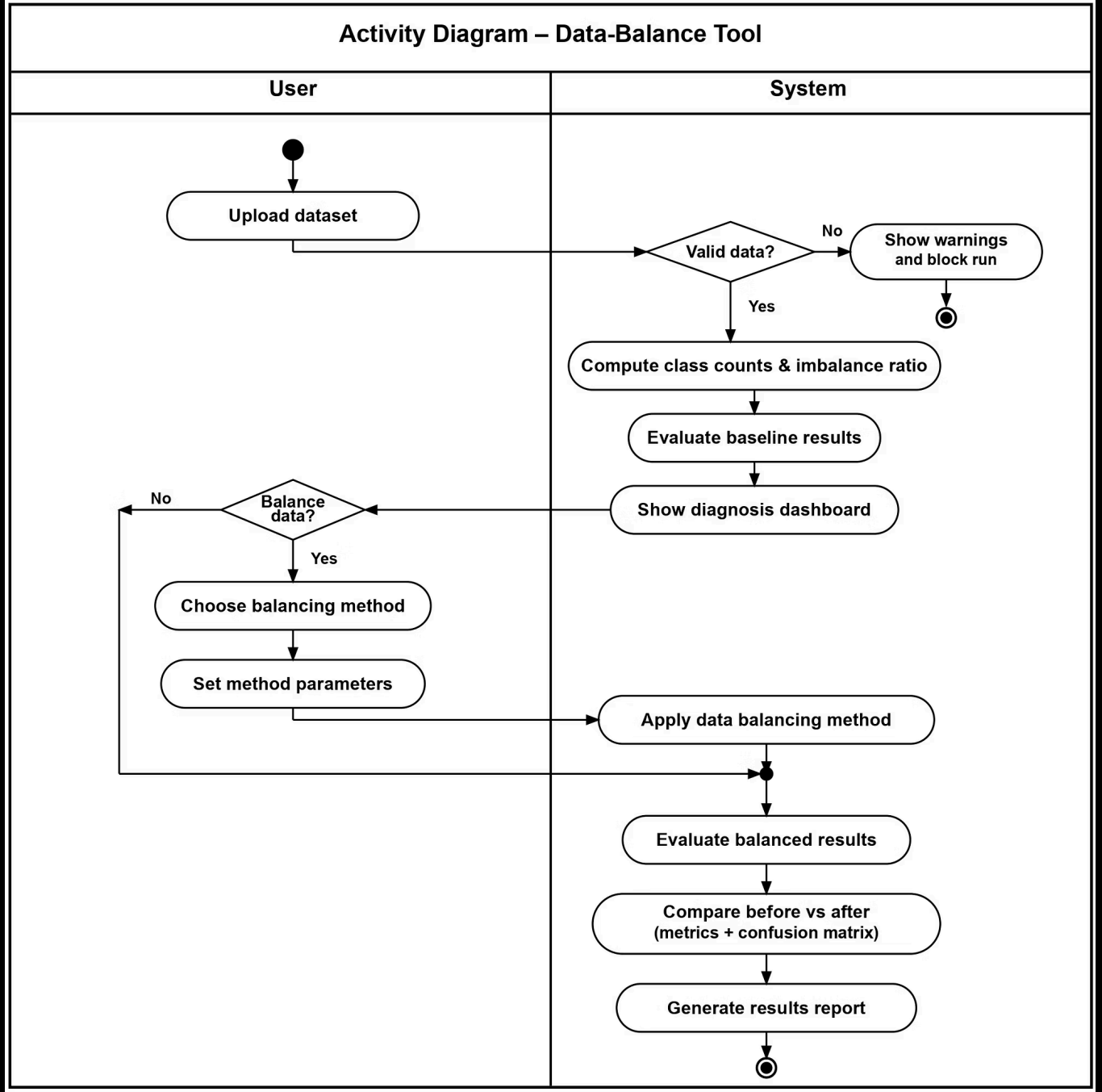
Next: who uses the system and what actions they perform (Use Case Diagram)

- Users: Researchers, Software Engineers, AI Developers.
- Core actions: Upload dataset, Analysis, Balancing method selection, Export.
- Outputs: Imbalance metrics + Evaluation, comparison report
- Goal: validate improvements without data leakage



Use Case Diagram




Activity Diagram – Data-Balance Tool
User
System
Upload dataset
Valid data?
No
Show warnings and block run
Yes
Compute class counts & imbalance ratio
Evaluate baseline results
Show diagnosis dashboard
Balance data?
No
Yes
Choose balancing method
Set method parameters
Apply data balancing method
Evaluate balanced results
Compare before vs after (metrics + confusion matrix)
Generate results report

Project Overview

Manage your dataset balancing projects

New Run

Upload a dataset and start a new balancing workflow

Start New Project

Open Previous Run

Continue working on a saved project or review past results

Browse Projects

Recent Runs

Your latest balancing projects

Dataset Name	Date	Method Used	Best Metric
credit_risk.csv	2026-01-05	SMOTE	0.89 F1
customer_churn.csv	2026-01-03	Hybrid	0.84 F1
fraud_detection.csv	2025-12-28	NearMiss	0.76 F1

Upload Dataset

Upload your CSV file and configure label settings

File Details

File name	Rows	Columns
credit_risk.csv	12,847	23

Replace File

Label Configuration

Select the target column for classification

Select label column

default

Choose the column that contains your classification target

Detected classes: 0 (non-default), 1 (default)

Validation Results

Dataset quality checks and warnings

- No missing values in label column — PASS
- Sufficient samples per class (min: 150) — PASS
- Severe class imbalance detected (IR: 8.4:1) — WARN
- 4 features contain missing values (5-12% per feature) — WARN

Evaluation

Compare model performance before and after balancing

Before Balancing

Classification Metrics

Accuracy	0.879
Precision	0.623
Recall	0.412
F1 Score	0.496
Macro F1	0.681

Confusion Matrix

	Pred 0	Pred 1
Actual 0	2,184	76
Actual 1	182	128

After Balancing

Classification Metrics

Accuracy	0.841
Precision	0.781
Recall	0.823
F1 Score	0.801
Macro F1	0.836

Confusion Matrix

	Pred 0	Pred 1
Actual 0	2,003	257
Actual 1	55	255

Performance Delta

Changes in key metrics after balancing

Accuracy	-0.038 ↓
Precision	+0.158 ↑
Recall	+0.411 ↑
F1 Score	+0.305 ↑
Macro F1	+0.155 ↑

Summary: Significant improvement in minority class detection (+41% recall) with strong F1 gain (+30%).

Evaluation & Success

Evaluation Approach

Software Tests

- Data validation
- Algorithm correctness
- Export integrity

Research Tests

- Fair comparisons
- Reproducibility

Key Performance Indicators

- **Runtime:** <60s for supported datasets
- **Imbalance reduction:** IR decreases after balancing
- **Minority-class recall:** Improves on most datasets
- **Macro-F1:** Increases or stable
- **Per-class reporting:** 100% Precision/Recall/F1
- **Export success:** 100% in normal usage

Success Criteria

1. Accurate diagnosis of imbalance level
2. Effective balancing improves class distribution
3. Clear before/after comparison shows balancing effect

Conclusions

- Class imbalance causes models to fail on critical minority cases despite high accuracy
- Accuracy alone is misleading—must use balanced metrics (G-Mean, per-class recall, Macro-F1)
- Our unified workflow enables reproducible diagnosis, handling, and comparison
- Transparent documentation ensures reliable methodology and fair evaluation

Thank You

Questions?

GitHub repository and appendices available. Contact information available upon request.